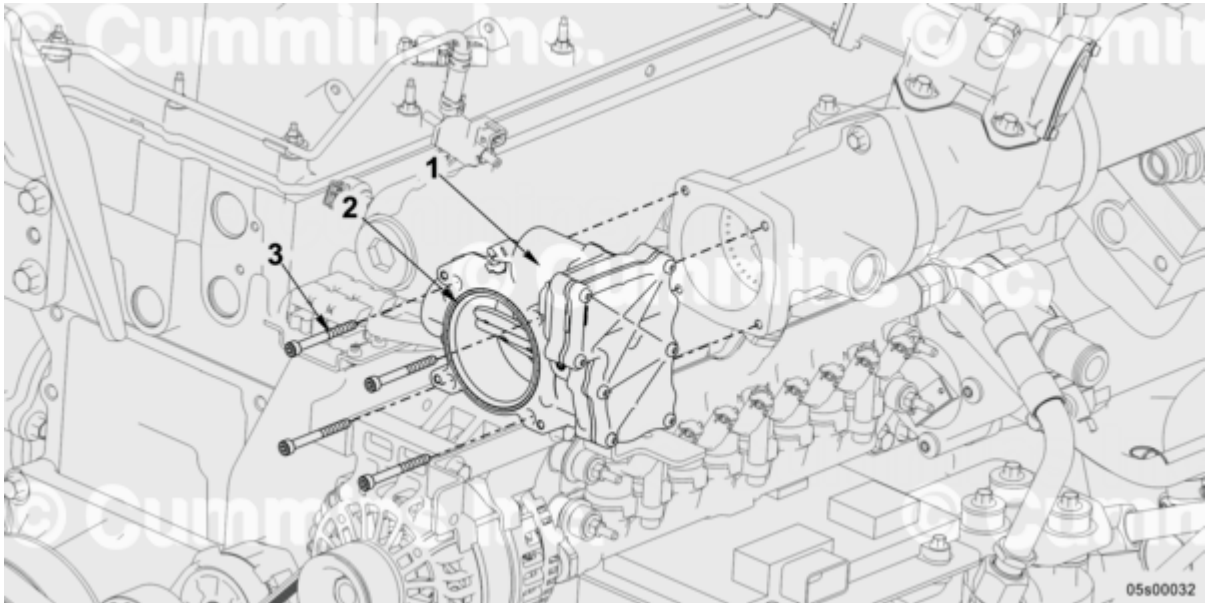


Trainings ▾

Exploded View



Throttle Body Assembly Exploded View

1. Throttle body assembly - quantity 1
2. O-ring - quantity 2
3. Capscrew - quantity 4

Select Service Tools

Recommended Cummins® Service Tools

- Cummins® electronic service tool, or equivalent

Additional Service Items

- Plastic or brass bottle brush.

Initial Check

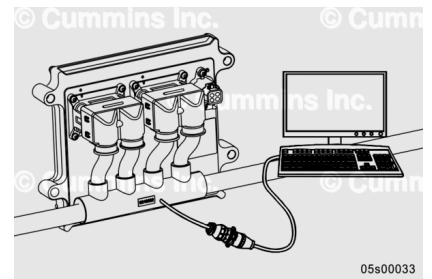
Use the recommended Cummins® electronic service tool, or equivalent, to perform the throttle diagnosis.

Turn the keyswitch to the ON position.

Connect the electronic service tool to the service tool data link, which is located on the engine or in the cab.

Open the throttle diagnosis interface in the electronic service tool.

Depress the accelerator pedal, and check if the readings of the accelerator pedal position and the throttle valve position in the electronic service tool are synchronous. If **not**, repair or replace the throttle body assembly.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

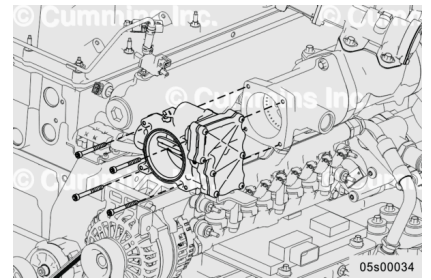
Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the air intake connection. Refer to Procedure 010-080 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-080.html>)
- Remove the air intake connection adapter. Refer to Procedure 010-131 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-131.html>)
- Disconnect the wiring harness connector from the throttle actuator. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html>)

Remove

Remove the engine throttle actuator.

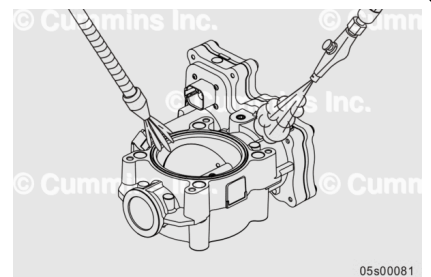
Discard the two o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Quick or forceful movements of the throttle plate by hand can result in damage to the mechanism.

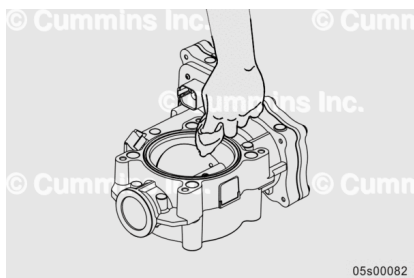
Clean the intake fitting adapter elbow and engine intake throttle actuator using solvent.

Do **not** immerse in solvent.

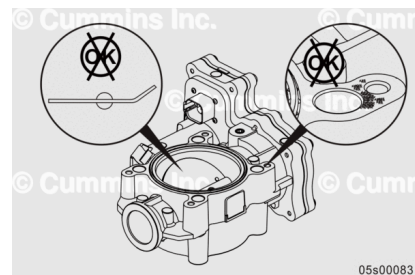
If the throttle body assembly has frozen, defrost the sensor using warm air that is below 60°C [140°F].

Dry with compressed air.

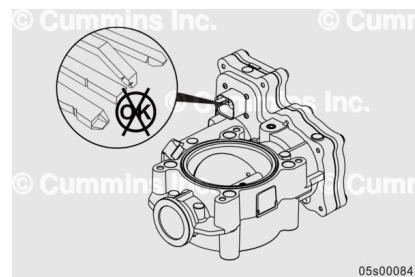
Clean the inside of the bore and plate with a clean lint-free cloth to remove any remaining contaminants.



Inspect the engine throttle actuator for a bent throttle plate or damage around the sealing surfaces.



Inspect the wiring connections for bent or damaged pins.



Install

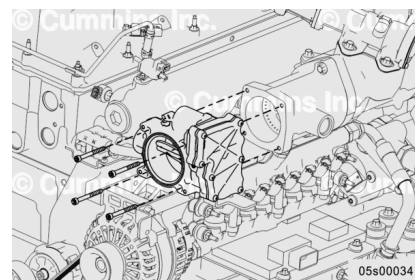
Lubricate the o-rings with a very light coat of vegetable oil.

Install o-rings on the engine throttle actuator.

Install the engine throttle actuator to the air/fuel mixer.

Tighten the four capscrews.

Torque Value: 18 n•m [159 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Natural gas is explosive and flammable. Always be sure to maintain adequate ventilation in the work area. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas with shared ventilation to reduce the possibility of severe personal injury or death when working on a natural gas system.

- Install the air intake connection adapter. Refer to Procedure 010-131 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-131.html>)
- Install the air intake connection. Refer to Procedure 010-080 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-080.html>)
- Connect the wiring harness connector with the throttle actuator. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html>)
- Connect the batteries. See equipment manufacturer service information.
- Start and check for proper operation.